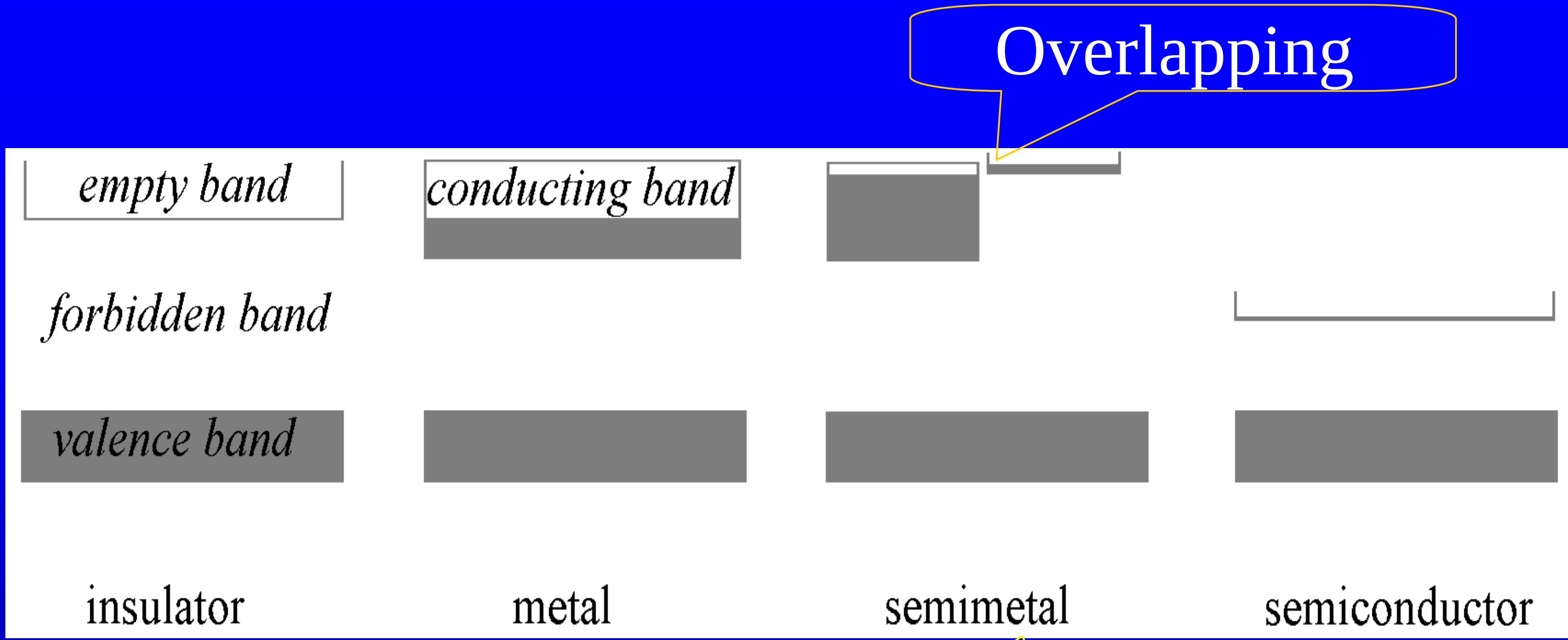
Chapter 16 Conductor and Dielectrics

Plenty of free electrons are the characteristic of a conductor.

The atoms consisting conductor usually have unfilled shell, e.g. $_{11}Na:1s^2 2s^2 2p^6 3s$.

When a crystal is formed, we have energy bands



e. g.,Alkaline earth metals
Mg (magnesium) $3s^2$

A complete filled band does not contribute current when the electric field is applied.

16.1 Uniform conductor in an electrostatic field

electrostatic equilibrium requires $E_{\text{in}} = 0$, $E_{//} = 0$(on the surface)

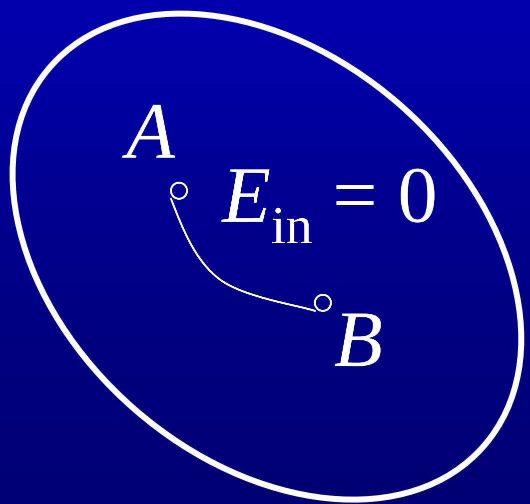
$$V_B - V_A = -\int_A^B \boldsymbol{E}_{//} \cdot d\boldsymbol{l} = 0$$

the surface of a conductor is an equipotential surface

$E_{\text{in}} = 0$ also means that the conductor is an equipotential body.

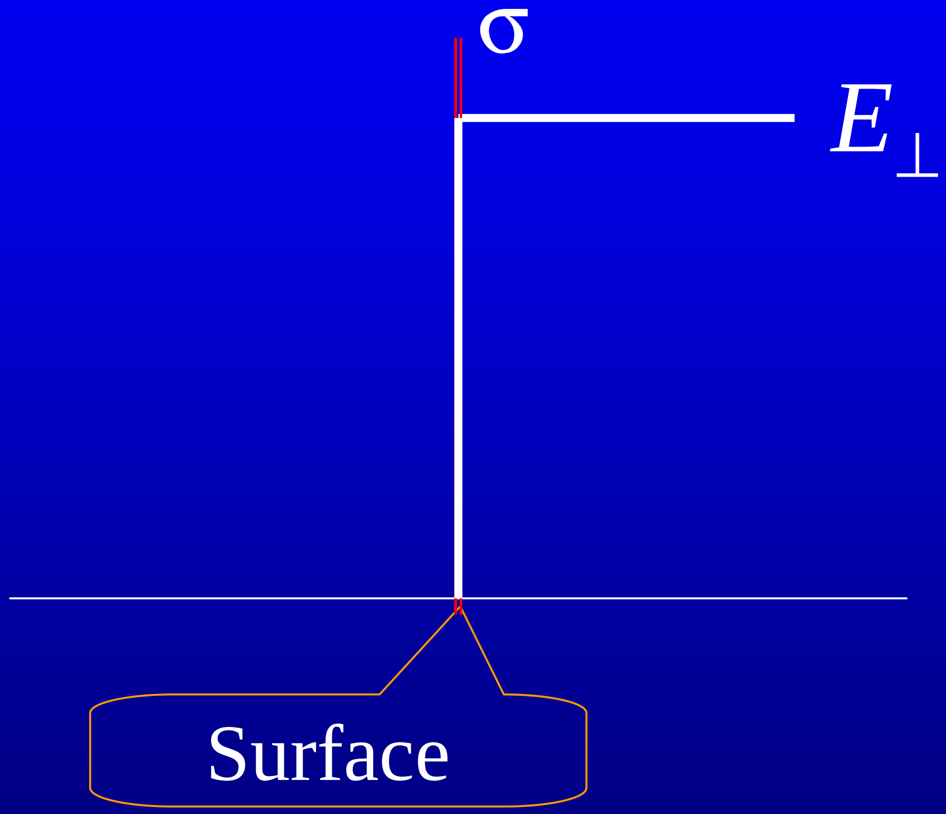
$$V_B - V_A = -\int_A^B \boldsymbol{E}_{\text{in}} \cdot d\boldsymbol{l} = 0$$

$$\nabla V = 0 \Rightarrow V = C.$$



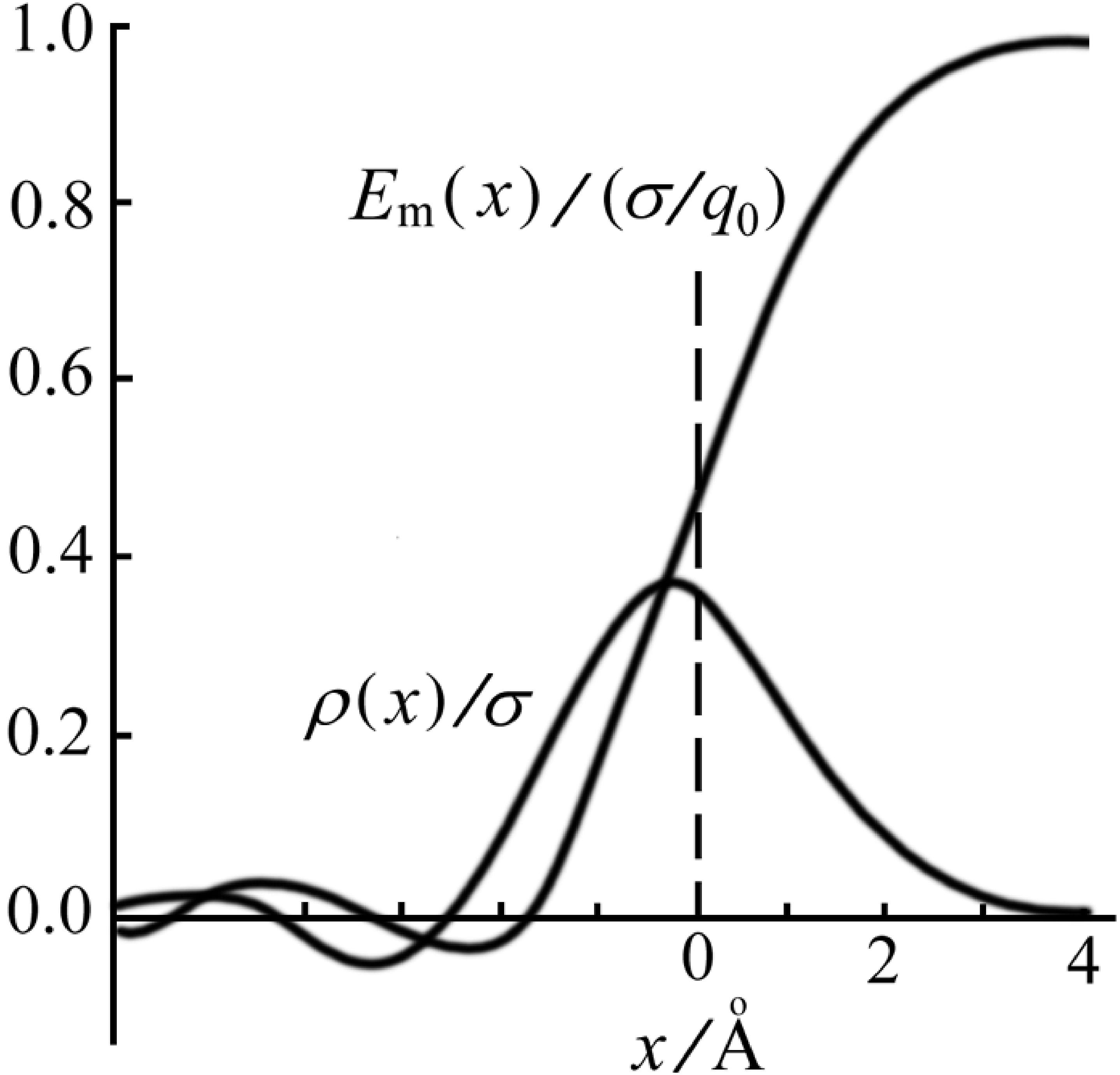
The field near the surface of the conductor

$$E_{\perp} = \frac{\sigma}{\varepsilon_0}$$



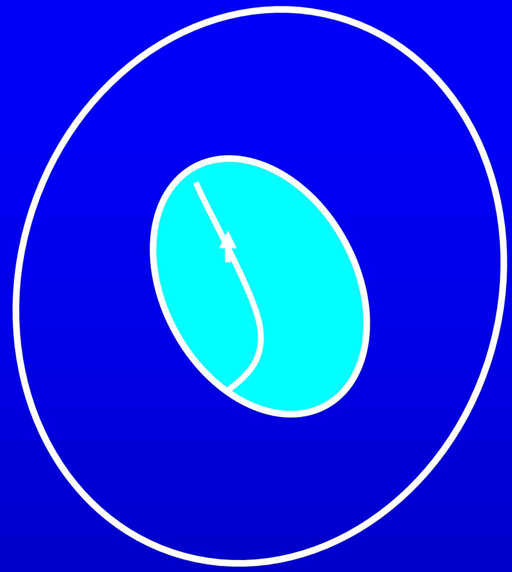
?

On microscopic scale, is it a real step function?



1.0
0.8
0.6
0.4
0.2
0.0
$E_m(x)/(\sigma/q_0)$
$\rho(x)/\sigma$
0
2
4
$x/\text{Å}$

A conductor with a cavity

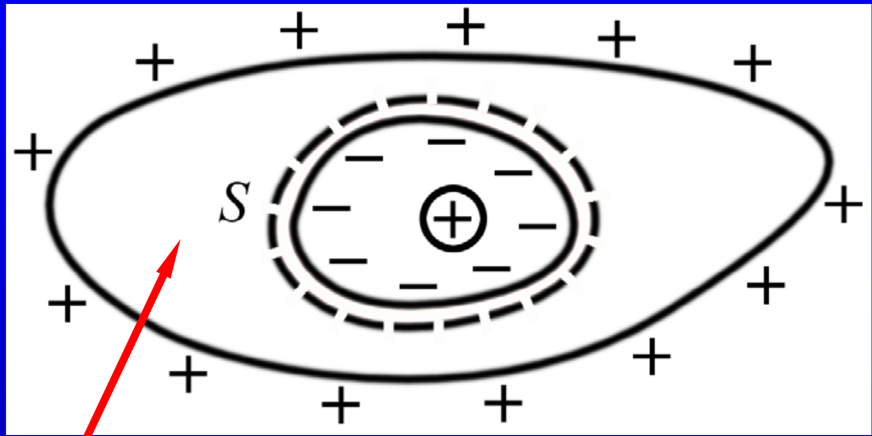


S-a closed surface just around the cavity

$$\oint_S \boldsymbol{E} \cdot d\boldsymbol{S} = 0 \Rightarrow$$ The total charge on the surface of the cavity is zero.

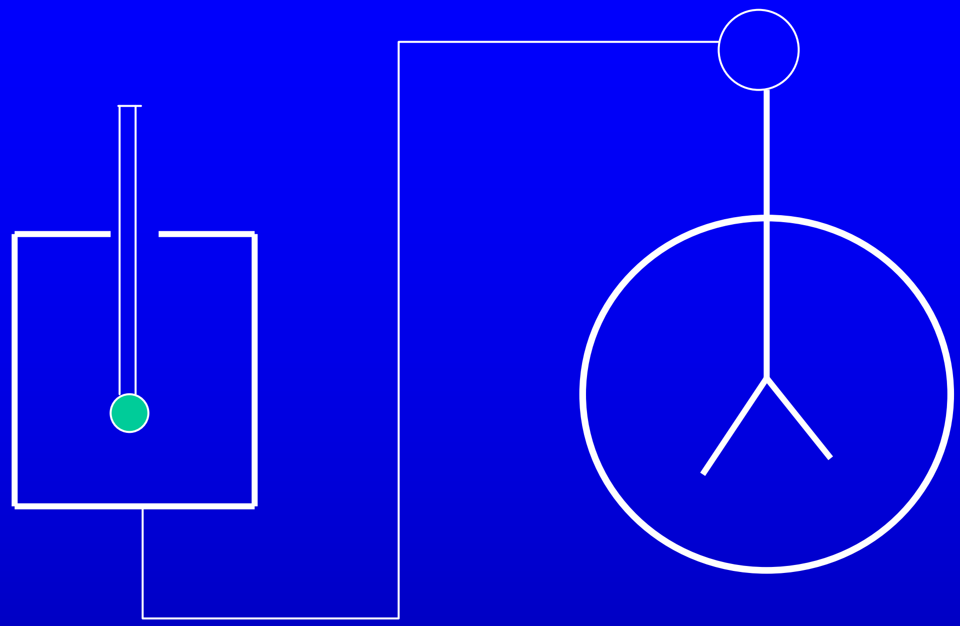
The charge density on each point of the inner surface is zero.

Electrostatic shielding



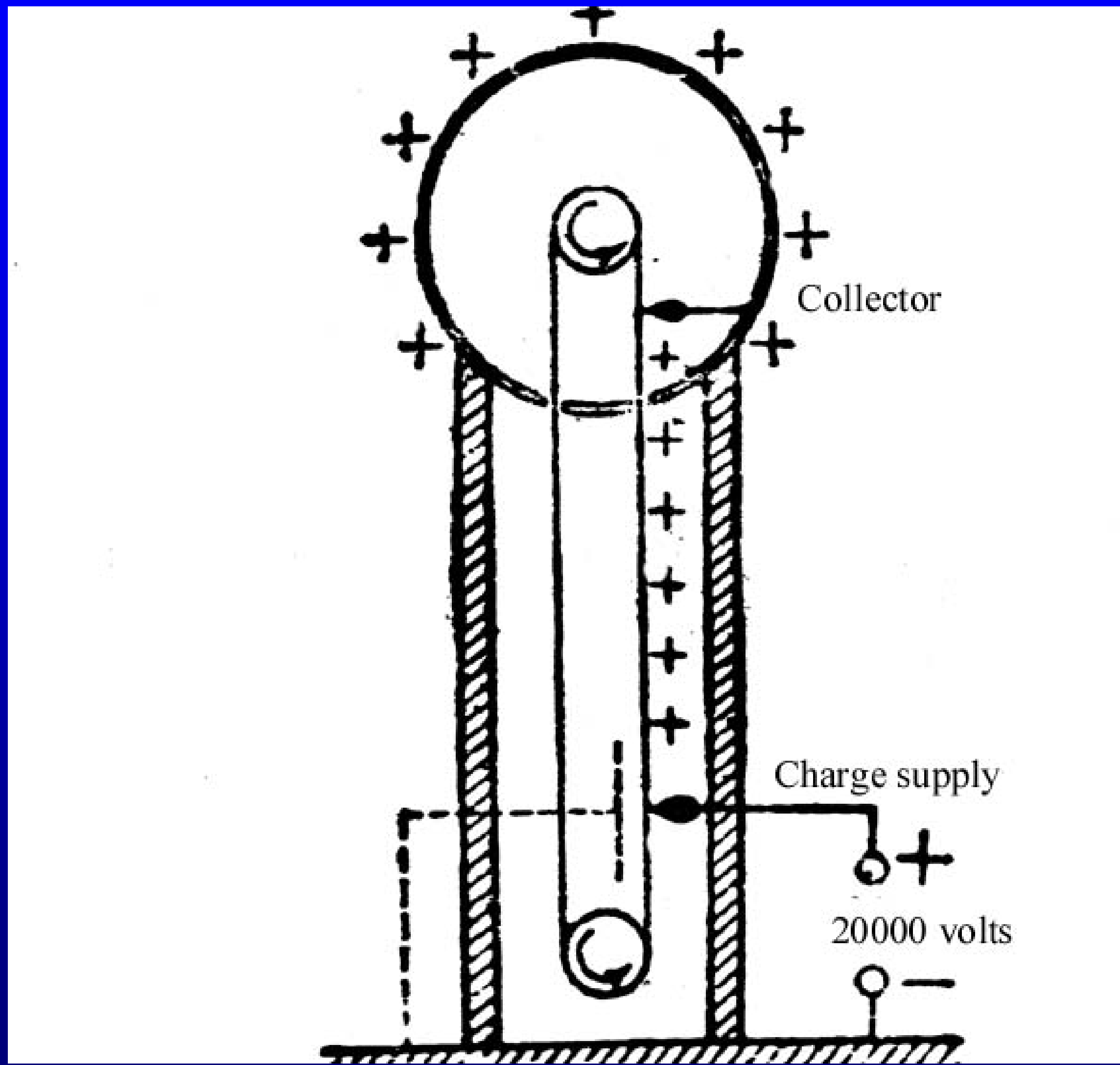
$\boldsymbol{E} = 0$

Faraday’s ice pail

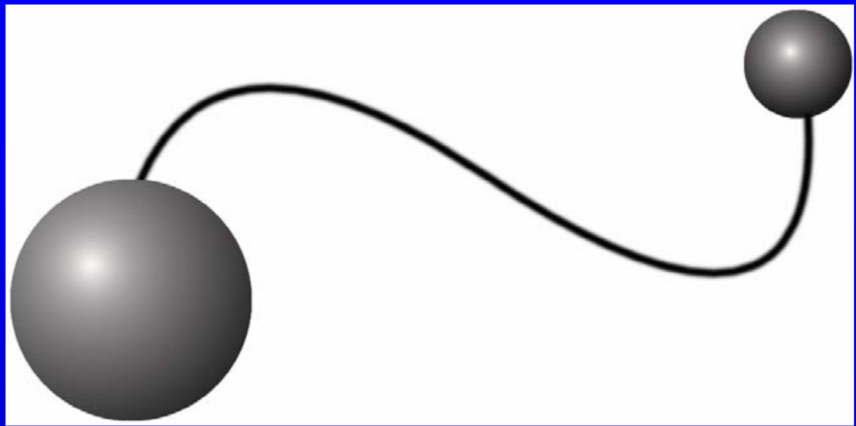


electroscope

Van de Graaf generator 10^7 volts



The charge density on the surface of a conductor



$$V_1 = \frac{1}{4\pi\varepsilon_0}\frac{q}{r} = V_2 = \frac{1}{4\pi\varepsilon_0}\frac{Q}{R}$$

$$\frac{\sigma_1}{\sigma_2} = \frac{q}{4\pi\, r^2} : \frac{Q}{4\pi\, R^2} = \frac{R}{r}$$

$$E \sim \frac{\sigma}{\varepsilon_0} \sim \frac{1}{\rho}$$

The surface charge density as well as the electric field is inversely proportional to the radius of curvature.

3×10^{6} N/C corona discharge lightning rod

16.2 Capacitance

capacitor

for a isolated conductor $$C = \frac{q}{V}$$

capacitance of a sphere or shell

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{R}, \quad C = \frac{q}{V} = 4\pi\varepsilon_0 R$$

units

coulombs/volt (C/V) farads (F)

microfarads (μ F) , picofarads (pF)

$$1\mu\mathrm{F} = 10^{-6}\,\mathrm{F}, \quad 1\mathrm{pF} = 10^{-12}\,\mathrm{F}$$

Capacitance of two conductor A, B

$$C_{AB} = \frac{q_A}{V_A - V_B}$$

In a parallel-plate capacitor

$$E = \frac{\sigma}{\varepsilon_0}$$

Ignore the edge effect

$$V_U - V_L = -\int \boldsymbol{E} \cdot d\boldsymbol{l} = Ed = \frac{\sigma}{\varepsilon_0} d$$

upper plate

lower plate

$$C = \frac{q}{\Delta V} = \frac{\sigma A}{\frac{\sigma}{\varepsilon_0} d} = \frac{\varepsilon_0 A}{d}$$

Example 16.1 Capacitance of concentric spherical shells of radii *R* and *R'*.

$$E_r = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2} \quad (R < r < R')$$

$$V_R - V_{R'} = -\int_{R'}^{R} \boldsymbol{E}\cdot d\boldsymbol{l} = \int_{R}^{R'} \boldsymbol{E}\cdot d\boldsymbol{r}$$

$$= \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{R} - \frac{1}{R'}\right)$$

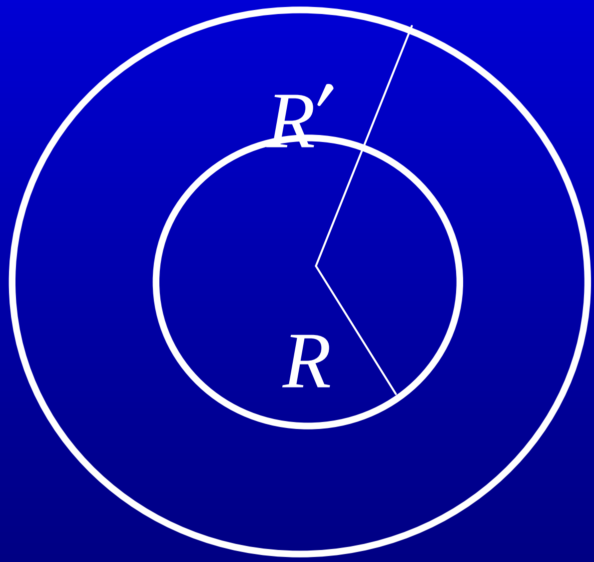
$$C = \frac{q}{V_R - V_{R'}} = 4\pi\varepsilon_0 \frac{RR'}{R' - R}$$

$$C = \frac{q}{V_R - V_{R'}} = 4\pi\varepsilon_0 \frac{RR'}{R' - R}$$

It can be considered as capacitors combined in series

$$\frac{1}{C} + \frac{1}{4\pi\varepsilon_0 R'} = \frac{1}{4\pi\varepsilon_0 R}$$

$$\frac{1}{C} = \frac{1}{4\pi\varepsilon_0 R} - \frac{1}{4\pi\varepsilon_0 R'}$$



When R and R' approach to the infinity and the difference R'-R keeps constant d, we have

$$C = 4\pi\varepsilon_0 \frac{R(R+d)}{d}$$

$$= \varepsilon_0 4\pi R^2 \frac{\left(1+\frac{d}{R}\right)}{d} \rightarrow \frac{\varepsilon_0 A}{d}$$

result for parallel-plate capacitor

Example Find the capacitance of a cylindrical capacitor.

Solution: Assume the radius of the inner conducting wire is *a* and the radius of the out cylindrical shell is *b*. Let *L* be the length of the capacitor. *Q* and -*Q* be the charges on the inner and outer conductors respectively. The electric field between two conductors is

$$E_r = \frac{1}{2\pi\varepsilon_0}\frac{\lambda}{r} = \frac{1}{2\pi\varepsilon_0}\frac{Q}{rL}.$$

coaxial cable

The potential difference between the two conductors is then

$$\Delta V = V_a - V_b = -\int_b^a E_r dr$$

$$= -\frac{Q}{2\pi\varepsilon_0 L}\int_a^b \frac{dr}{r} = \frac{Q}{2\pi\varepsilon_0 L}\ln\frac{b}{a},$$

and the capacitance is

$$C = \frac{Q}{\Delta V} = \frac{2\pi\varepsilon_0 L}{\ln b/a}.$$

Suppose that battery does work and move dq from one plate to another

$$\mathrm{d}W = \Delta V \mathrm{d}q = \frac{q}{C}\mathrm{d}q$$

$$W = \int \mathrm{d}W$$

$$= \frac{1}{2C}q^2 = \frac{1}{2}C(\Delta V)^2 \equiv U$$

$$u = \frac{U}{Ad} = \frac{\frac{1}{2}C(\Delta V)^2}{Ad} = \frac{\frac{1}{2}\frac{\varepsilon_0 A}{d}(\Delta V)^2}{Ad}$$

$$= \frac{\varepsilon_0}{2}\left(\frac{\Delta V}{d}\right)^2 = \frac{1}{2}\varepsilon_0 E^2$$

* Forces on plates

$\boxed{q = \text{const.}}$

$$F_{\text{ext}}\,dx = -F_x\,dx = dU$$

$$F_x = -\left(\frac{\partial U}{\partial x}\right)_q$$

For parallel plate capacitor

$$U = \frac{1}{2C}q^2 = \frac{x}{2\varepsilon_0 A}q^2$$

$$F_x = -\left(\frac{\partial U}{\partial x}\right)_q = -\frac{1}{2\varepsilon_0 A}q^2$$

$$\boxed{\Delta V = \text{const.}}$$

$$-F_x \; dx + dW^B = dU$$

$$dW^B = \Delta V dq = d\left(q\Delta V\right) = d\left[C\left(\Delta V\right)^2\right] = 2dU$$

$$F_x \; dx = dU$$

$$F_x = \left(\frac{\partial U}{\partial x}\right)_{\Delta V}$$

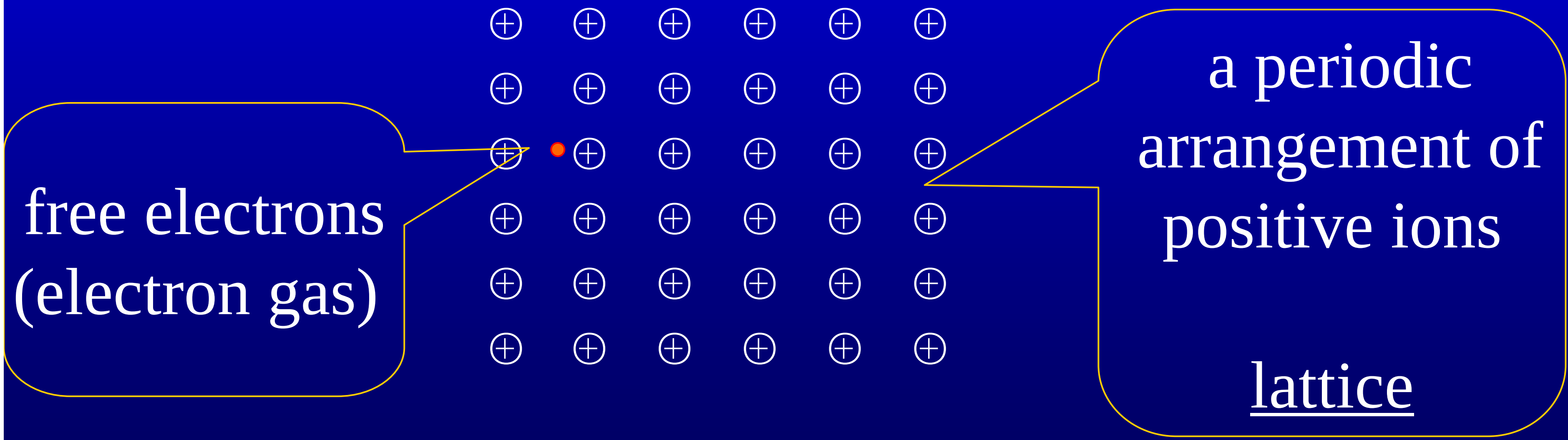
$$U = \frac{1}{2}C(\Delta V)^2 = \frac{1}{2}\frac{\varepsilon_0 A}{x}(\Delta V)^2 \qquad F_x = -\frac{\varepsilon_0 A}{2x^2}(\Delta V)^2$$

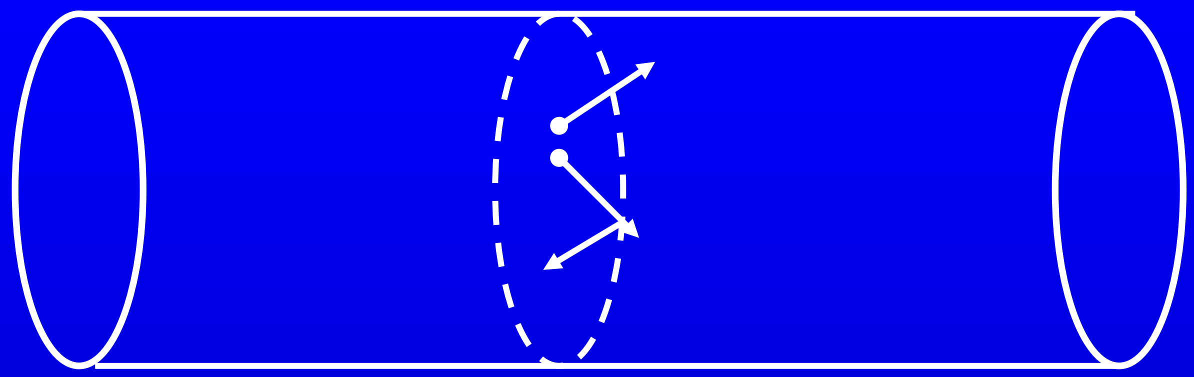
Assignment: 16.1, 16.3, 16.6,

16.3 Conductivity and Ohm's law

conductors	**charge carries**
metals	electrons
electrolytes	ions
plasmas	ions and electrons

Free electron model of a metallic conductor





When no external electric filed is present, the electrons move in all directions, no net current results.

If an external field is applied, a drift motion appears, and an net electric current results.

thermal motion

$$\boldsymbol{j} = n(-e)\langle \boldsymbol{v} \rangle$$

drift velocity of electrons

$$\langle \boldsymbol{v} \rangle = \langle \boldsymbol{v}(k_{\mathrm{B}}T) + \boldsymbol{v}_d(\boldsymbol{E}) \rangle = \boldsymbol{v}_d(\boldsymbol{E}) = \frac{-e\boldsymbol{E}}{m}\tau$$

where $\tau = \lambda / v_{\mathrm{rms}}$ is the relaxation time. Then the electric current density is

$$\boldsymbol{j} = \frac{ne^2\tau}{m}\boldsymbol{E} = \sigma\,\boldsymbol{E}$$

Ohm's law in differential from

where σ is called electric conductivity.

unit : $\Omega^{-1}\mathrm{m}^{-1}$.

The electric conductivity for some typical substances

substances	conductivity ($\Omega^{-1}\cdot m^{-1}$)
silver	6.2×10^{7}
copper	5.8×10^{7}
water	2×10^{-4}
fused quartz	10^{-17}

The reciprocal of the conductivity σ is called the resistivity

$$\rho = \frac{1}{\sigma}.$$

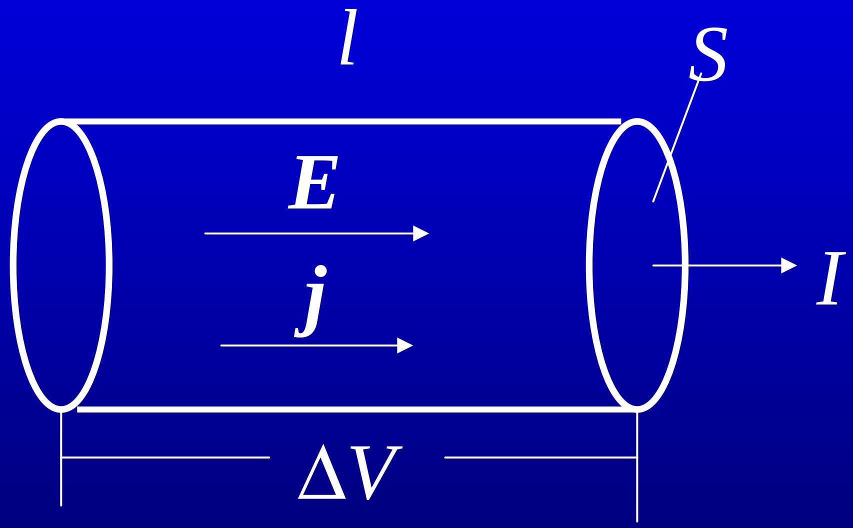
$$E = \frac{\Delta V}{l} \qquad j = \frac{I}{S}$$

$$\Downarrow$$

$$\frac{I}{S} = \sigma \frac{\Delta V}{l}$$

$$\Downarrow$$

$$I = \frac{\Delta V}{l/\sigma S} = \frac{\Delta V}{l\rho/S} = \frac{\Delta V}{R}$$



resistance

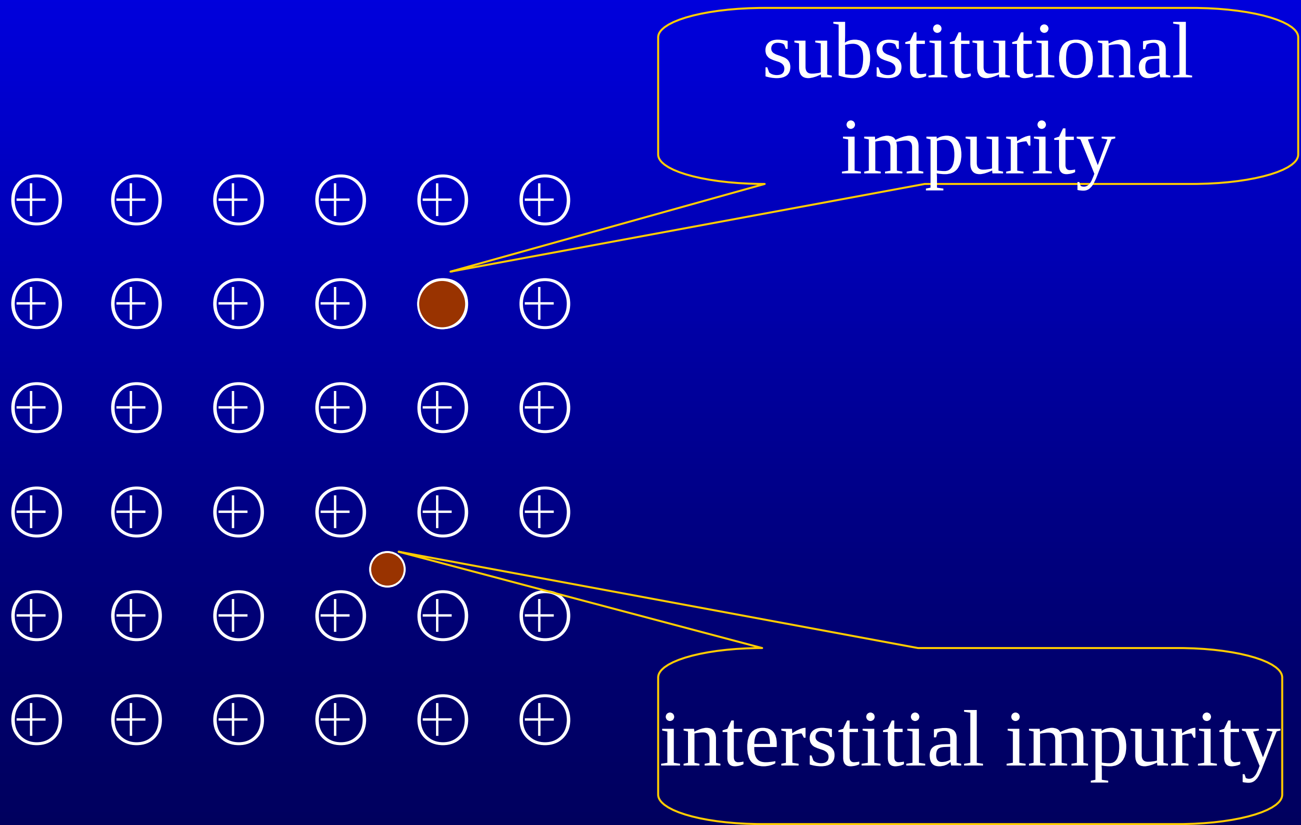
Ohm's law

The origin of the resistance

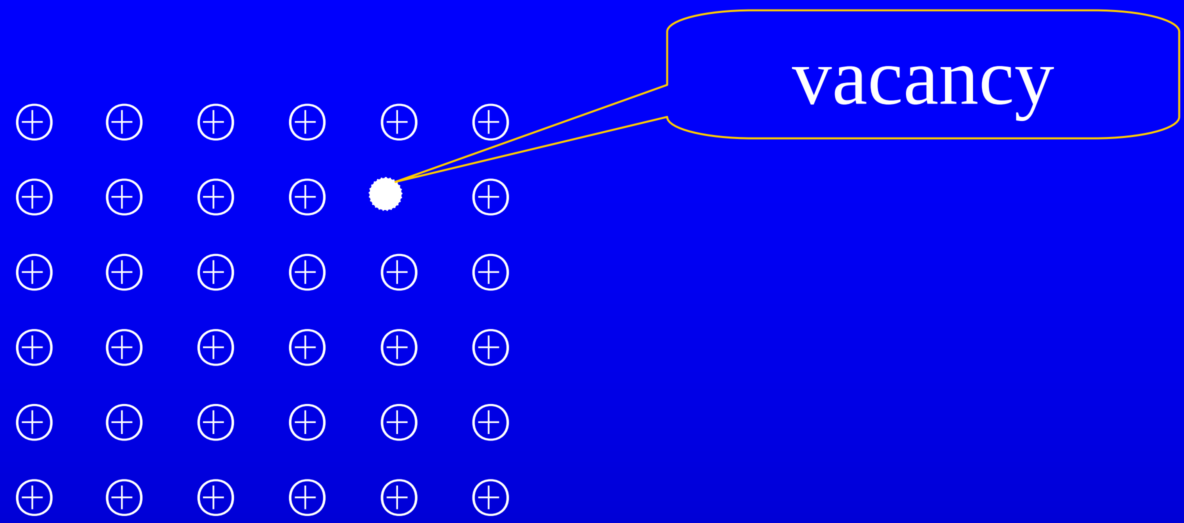
a perfect crystal lattice --- no resistance

a. imperfections of crystals

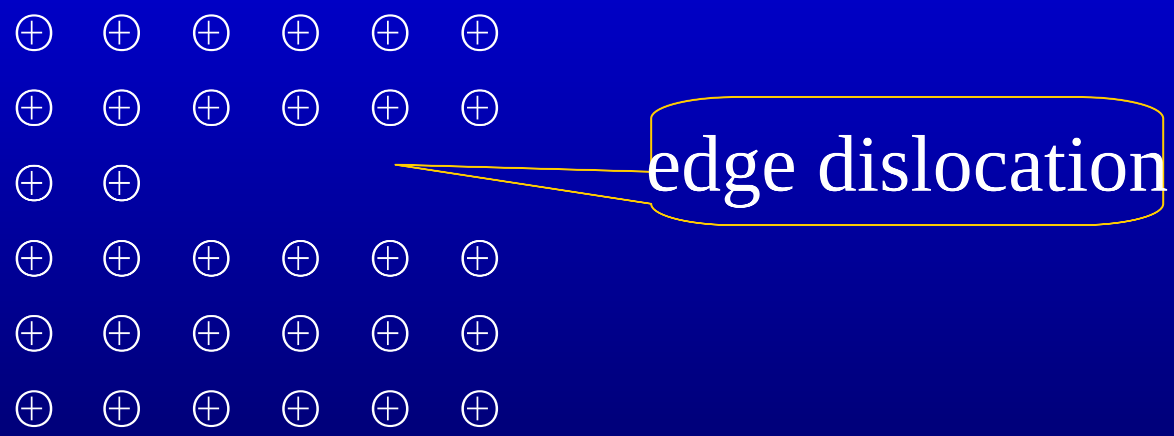
<u>impurities</u>



vacancies



dislocations



b. thermodynamic oscillations of the ions

Example Assume the current density in a copper wire is 2.4A/mm^2, the free electron density in the copper is $n = 8.4\times10^{28}\,/\,\mathrm{m}^2$. Estimate the drift velocity of the free electrons.

Solution:

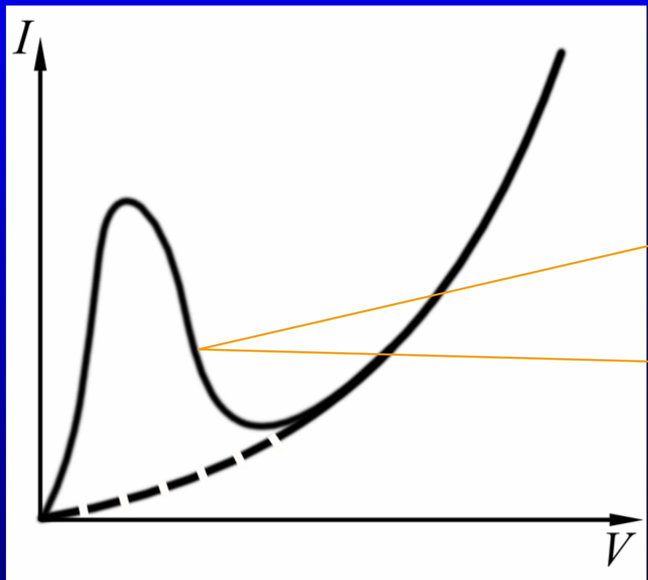
$$\upsilon_d = \frac{j}{ne} = \frac{2.4\times10^{6}\ \mathrm{A/\,m^2}}{8.4\times10^{28}\,/\,\mathrm{m}^3\times1.6\times10^{-19}\ \mathrm{C}}$$

$$= 1.8\times10^{-4}\ \mathrm{m/\,sec}.$$

Nonohmic conductors

$$I = \frac{V}{R} = GV, \quad G = R^{-1} \rightarrow \text{ conductance}$$

unit: Ω^{-1} or siemens (S)



negative differential conductance (resistance)

$G=dI/dV$

($R=dV/dI$)

Nonlinear voltage-current(V-A) relation of a tunnel diode

1900 Drude's discovery

$$j_e = -\sigma \nabla V$$

$$j_Q = -\kappa \nabla T$$

thermal conductivity

$$C_V = \frac{3}{2}\frac{n}{N_\mathrm{A}} R$$

$$= \frac{3}{2} n k_\mathrm{B}$$

$$\kappa = \frac{1}{3} v_\mathrm{rms} \bar{\lambda}\, n \left(\frac{3}{2} k_\mathrm{B} \right)$$

$$\sigma = \frac{ne^2\tau}{m} \approx \frac{ne^2}{m}\frac{\bar{\lambda}}{v_\mathrm{rms}}$$

$$\frac{\kappa}{\sigma} = \frac{1}{2} m v_{\mathrm{rms}}^2 \frac{k_{\mathrm{B}}}{e^2}$$

$$= \frac{3}{2} k_{\mathrm{B}} T \frac{k_{\mathrm{B}}}{e^2}$$

or

$$\frac{\kappa}{\sigma T} = \frac{3}{2}\left(\frac{k_{\mathrm{B}}}{e}\right)^2 = 1.11 \times 10^{-8}\ \mathrm{W} \cdot \Omega / \mathrm{K}^2$$

Correct result is

Wiedemann-Franz law

$$\frac{\kappa}{\sigma T} = \frac{\pi^2}{3}\left(\frac{k_{\mathrm{B}}}{e}\right)^2 = 2.43 \times 10^{-8}\ \mathrm{W} \cdot \Omega / \mathrm{K}^2$$

Electromotive force

$$\Delta V = \int_a^b \boldsymbol{E} \cdot d\boldsymbol{l} = RI.$$

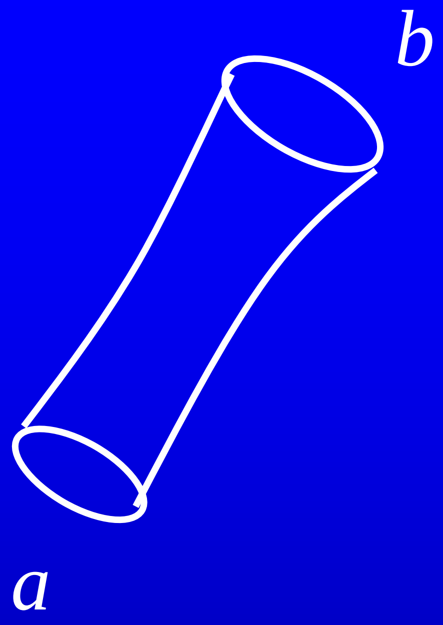
R-- resistance between *a* and *b*

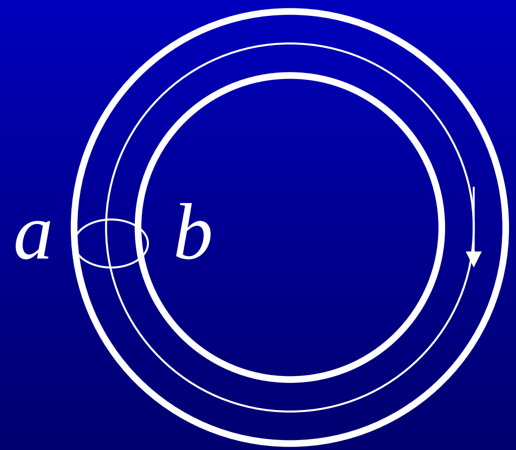
For a closed circuit, the above equation becomes

$$\oint \boldsymbol{E} \cdot d\boldsymbol{l} = RI.$$

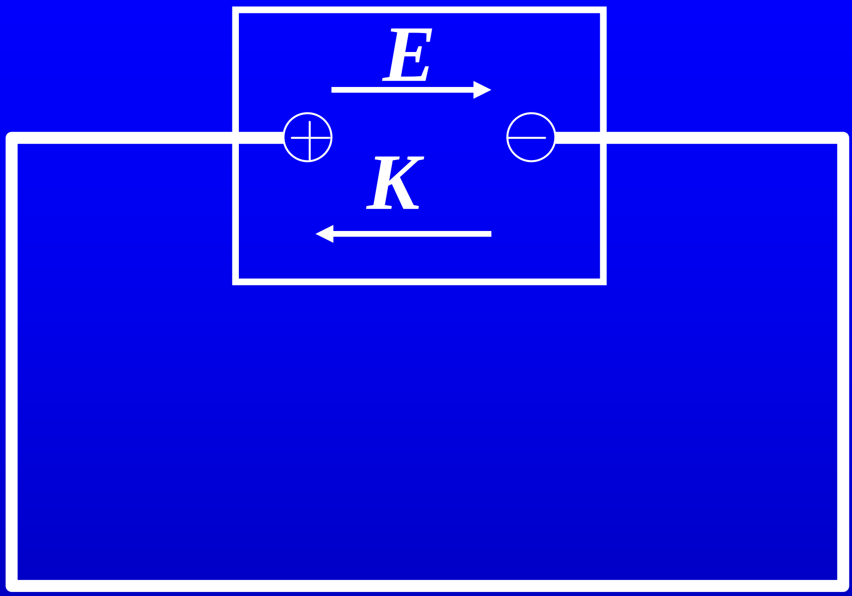
Since $\oint \boldsymbol{E} \cdot d\boldsymbol{l} = 0$ for static field, we have *I*=0.

<u>A static field cannot maintain a current in a closed circuit.</u>





electric power supplies: dry cell, storage battery, generator.



K--non-electrostatic field

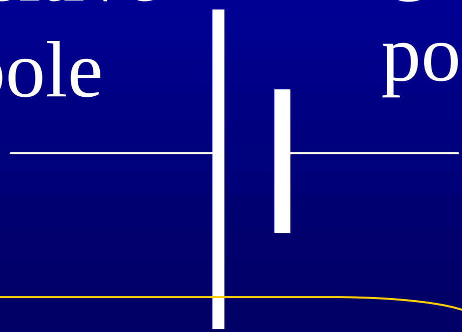
Ohm's law: $\boldsymbol{j} = \sigma(\boldsymbol{K} + \boldsymbol{E})$

The electromotive force can be defined as

$$\mathcal{E} = \int_{-}^{+} \boldsymbol{K} \cdot d\boldsymbol{l}$$

positive pole

negative pole



inside the power supply

For a closed circuit, we have

$$\oint_L (\boldsymbol{K}+\boldsymbol{E})\cdot d\boldsymbol{l} = IR.$$

or $$\oint_L \boldsymbol{K}\cdot d\boldsymbol{l} = IR \longrightarrow \mathcal{E} = IR.$$

For a open circuit, we have $\boldsymbol{j}$=0. Then

$$\boldsymbol{K} = -\boldsymbol{E} \longrightarrow \int_-^+ \boldsymbol{K}\cdot d\boldsymbol{l} = -\int_-^+ \boldsymbol{E}\cdot d\boldsymbol{l}.$$

$$\Big\downarrow$$

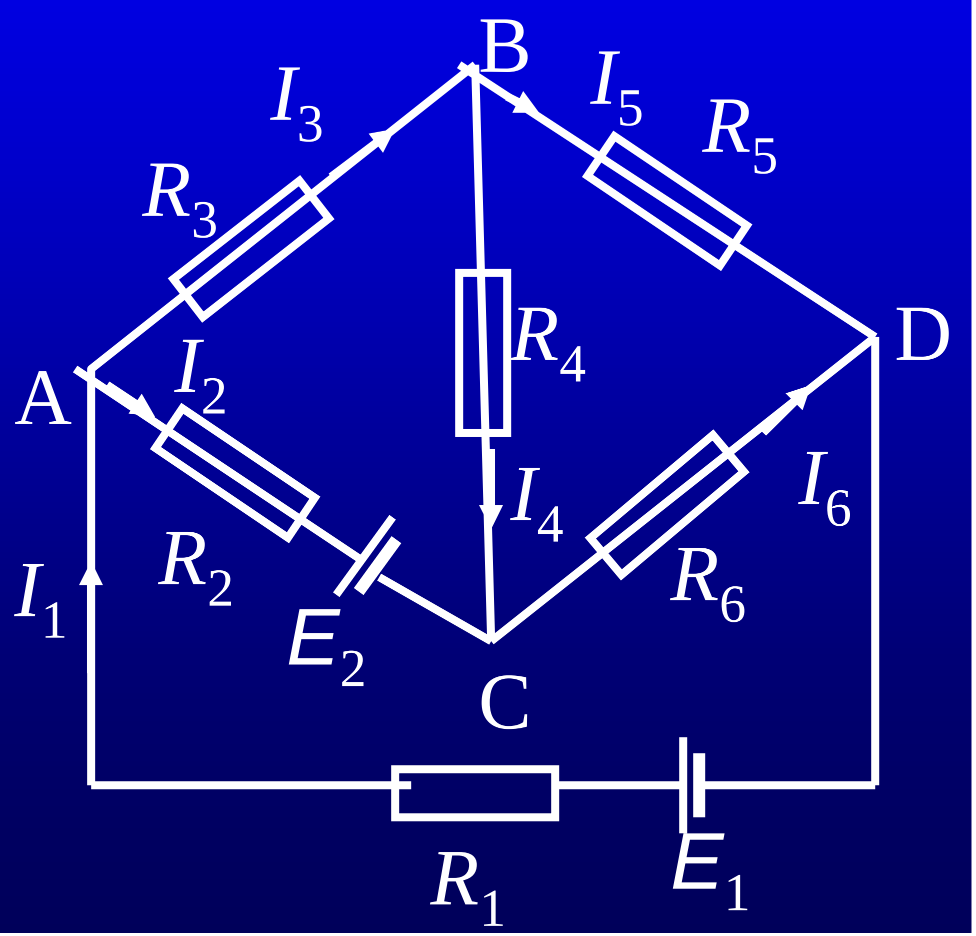
$$\mathcal{E} = \Delta V_{+\ -}.$$

Kirchhoff's rules

To calculate the currents in the complex circuit as shown in the figure, we usually apply Kirchhoff's rules.

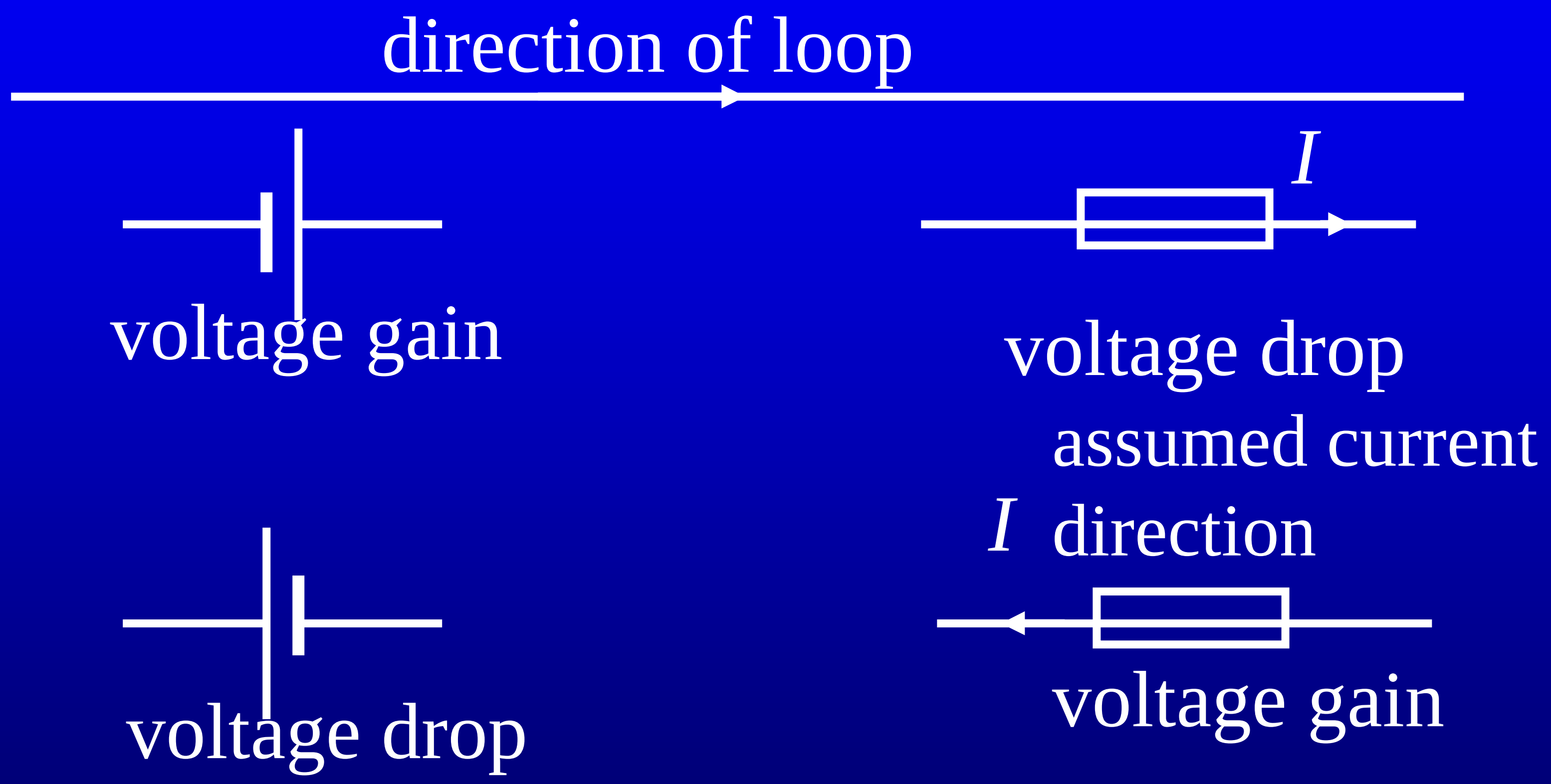
Loop rule: The algebraic sum of the voltage gains around any closed loop is zero.

Junction rule: The algebraic sum of currents flowing into any junction is zero.



Loop rule-- conservation of energy.

Junction rule-- conservation of charge.



<u>A voltage drop is a negative voltage gain.</u>

Loop 1: $\mathcal{E}_2 + I_2R_2 - I_3R_3 - I_4R_4 = 0;$

Loop 2: $I_4R_4 - I_5R_5 + I_6R_6 = 0;$

Loop 3: $\mathcal{E}_1 - I_1R_1 - I_2R_2 - \mathcal{E}_2 - I_6R_6 = 0;$

Junction A:

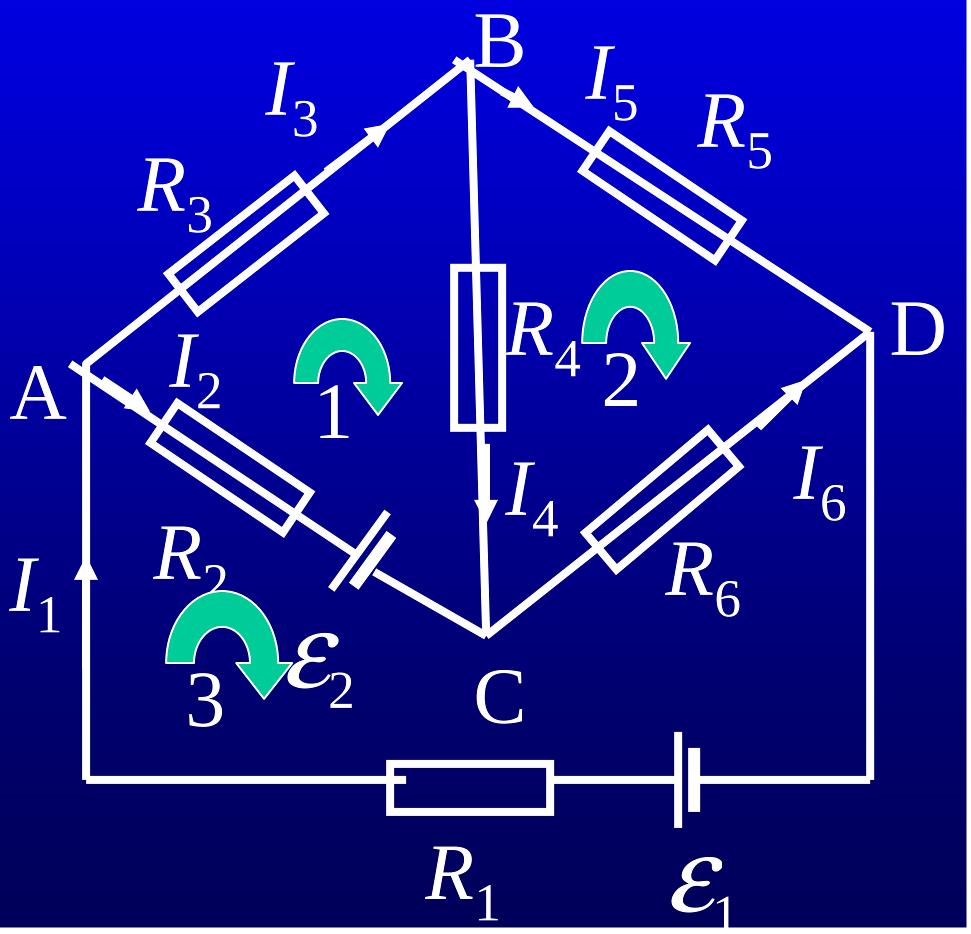
$$I_1 - I_2 - I_3 = 0;$$

Junction B:

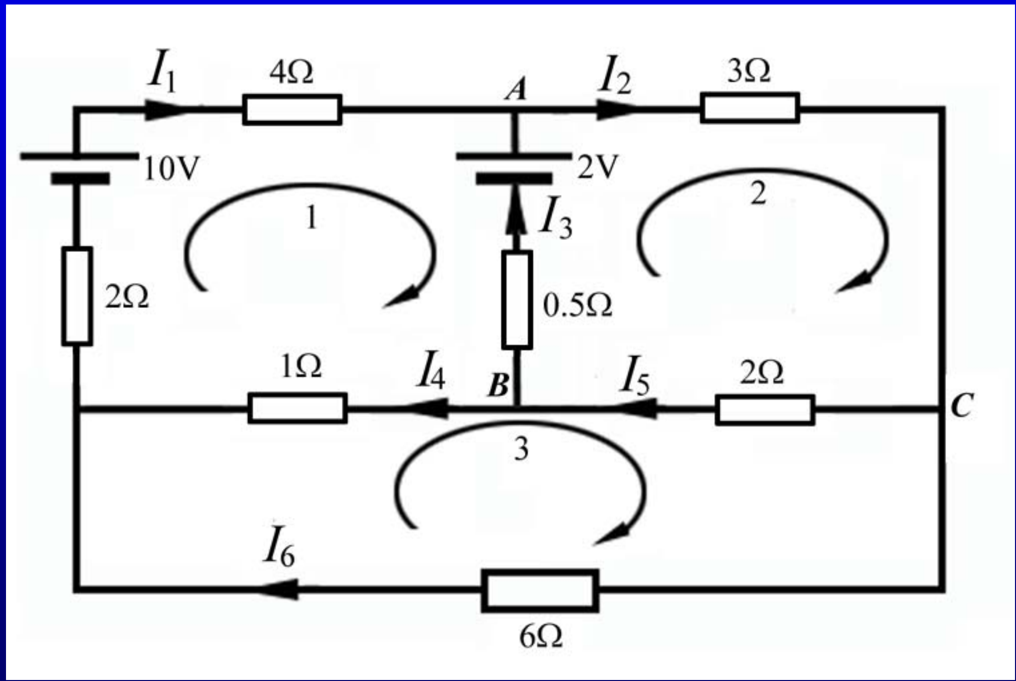
$$I_3 - I_4 - I_5 = 0;$$

Junction C:

$$I_2 + I_4 - I_6 = 0;$$



Supplementary problem 1. Using Kirchhoff's rules, determine the currents in the branches of the circuit shown in the accompanying figure.



16.4 Dielectrics

A nonconducting material , such as paper, wood, air, fused quartz, Teflon, amber and mica, is called dielectrics.

polar dielectrics	molecules with permanent dipole moment.
non-polar dielectrics	molecules without permanent dipole moment.
ferroelectrics	having a natural tendency for the permanent dipoles to align.

The electric polarization (vector)

$$\boldsymbol{P} = \lim_{\Delta V \to 0} \frac{(\Sigma \boldsymbol{p})}{\Delta V} = n \langle \boldsymbol{p} \rangle$$

total dipole moment per unit volume.

number of molecules per unit volume

average dipole moment of the molecules

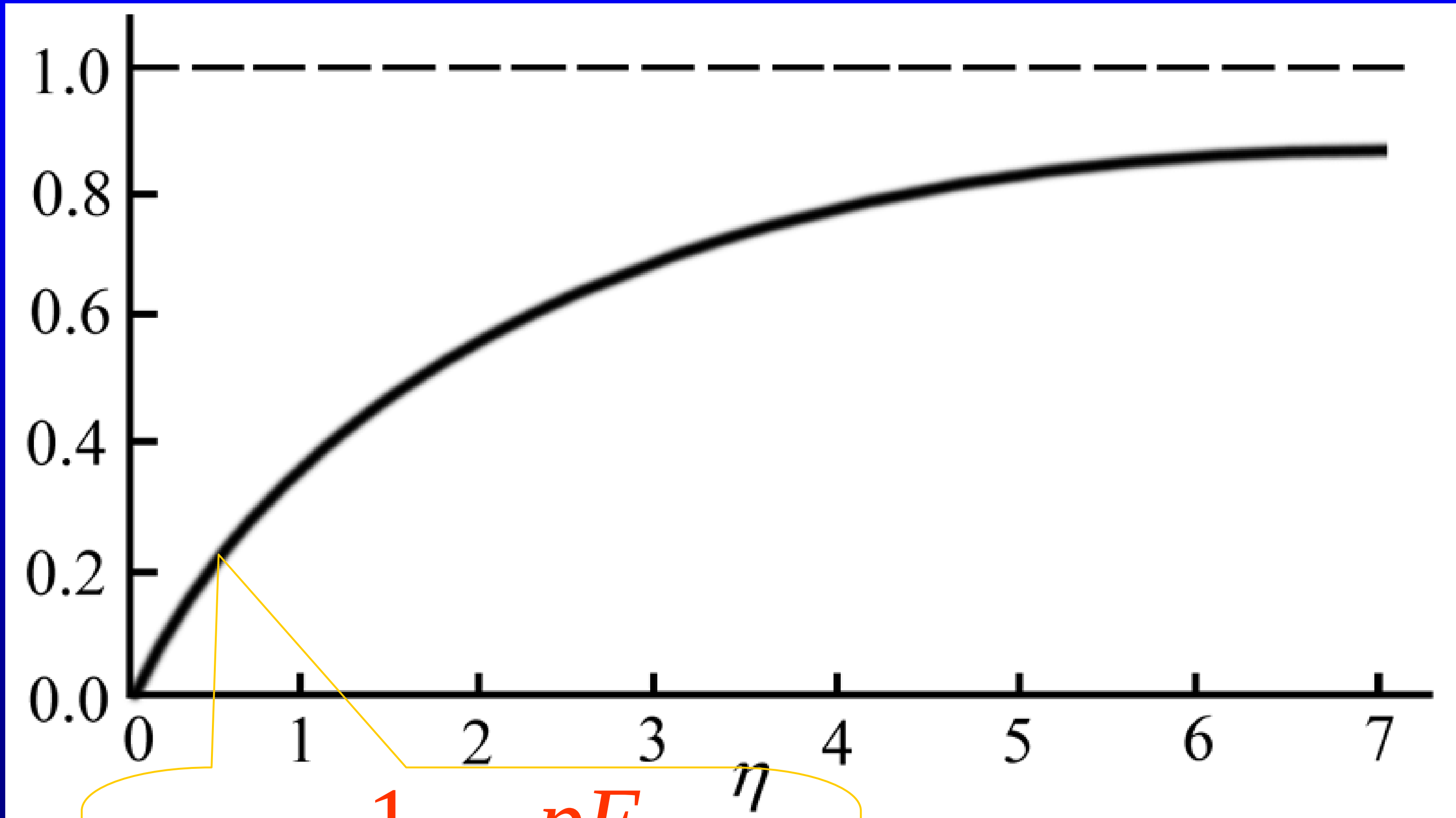
The permanent dipole moments of some molecules

H_2O,	$6.03\times10^{-30}\ \mathrm{C\cdot m}$
CO,	$4.00\times10^{-30}\ \mathrm{C\cdot m}$
HCl,	$3.43\times10^{-30}\ \mathrm{C\cdot m}$

The potential energy of a dipole in the field:

$$U = -\boldsymbol{p}\cdot\boldsymbol{E} + C = pE\cos\theta + C$$

Using Boltzmann's distribution, the average dipole moment in the field is



$$\langle \boldsymbol{p} \rangle = \frac{1}{3} p \frac{pE}{k_B T} \boldsymbol{k}$$

electric susceptibility

$$P = \chi_e \varepsilon_0 E$$

Molecules in non-polar dielectrics have no permanent dipole

Under the action of applied field they form induced dipole via distortion and displacement of the electron clouds relative to the atomic nuclei

$$\begin{aligned} \boldsymbol{p} &= 4\pi\varepsilon_0 R^3 \boldsymbol{E}_0 \\ &= \alpha \boldsymbol{E}_0 \end{aligned}$$

(Example 15.11)

gravitational or inertial forces, and mechanical stress (piezoelectric) may induce moments too

Ferroelectrics have spontaneous polarization

Barium titanate $BaTiO_3$

16.5 Electric Displacement Vector

Faraday found that when dielectric is inserted between two plates of the capacitor, the capacitance increased. The ratio of new capacitance C_d to C is

$$\frac{C_{\mathrm{d}}}{C} = \frac{q_{\mathrm{d}}}{q} \equiv \varepsilon_{\mathrm{r}} \geq 1 \qquad (\Delta V = \text{const.})$$

$$\frac{C_{\mathrm{d}}}{C} = \frac{\Delta V}{\Delta V_{\mathrm{d}}} = \varepsilon_{\mathrm{r}} \qquad (q = \text{const.})$$

Then, the capacitance of the parallel- plate capacitor with dielectric is

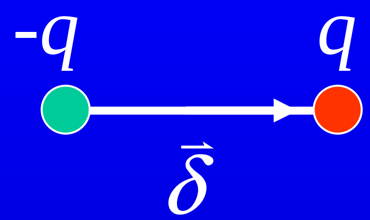
$$C_d = \varepsilon_r C = \frac{\varepsilon_r \varepsilon_0 A}{d} = \frac{\varepsilon A}{d}.$$

ε ---electric permittivity, dielectric constant

ε_r ---relative permittivity, relative dielectric constant

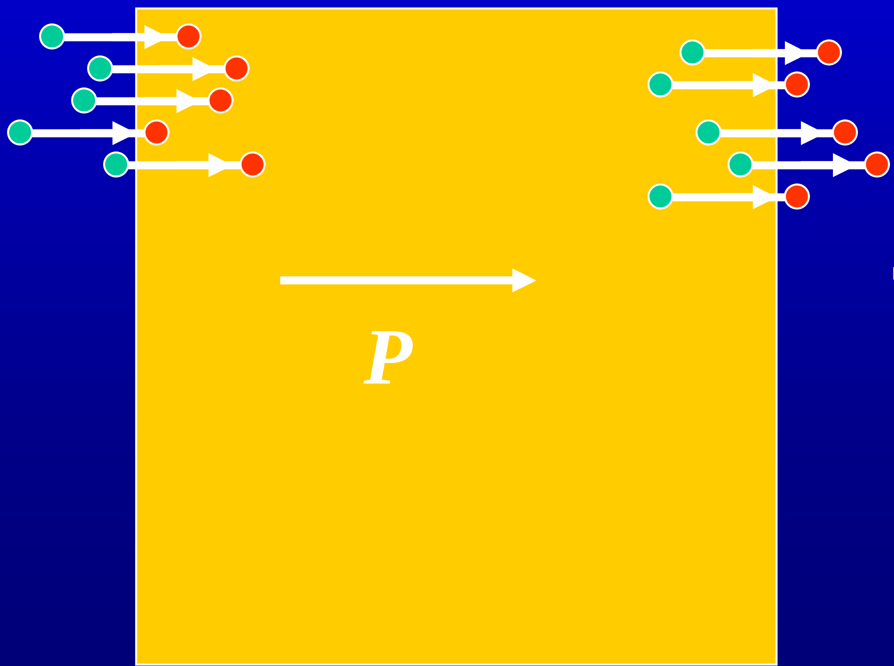
The increase of the capacitance is due to the polarization charge.

Model: For each molecule there are charges $\pm q$ separated by a distance $\vec{\delta}$.



The electric polarization vector is $\boldsymbol{P} = nq\vec{\delta}$.

number of molecules per unit volume



The polarization charge surface densities are

$$\pm\sigma_P = \pm n\delta q = \pm P.$$

Thus, the field between two plates is

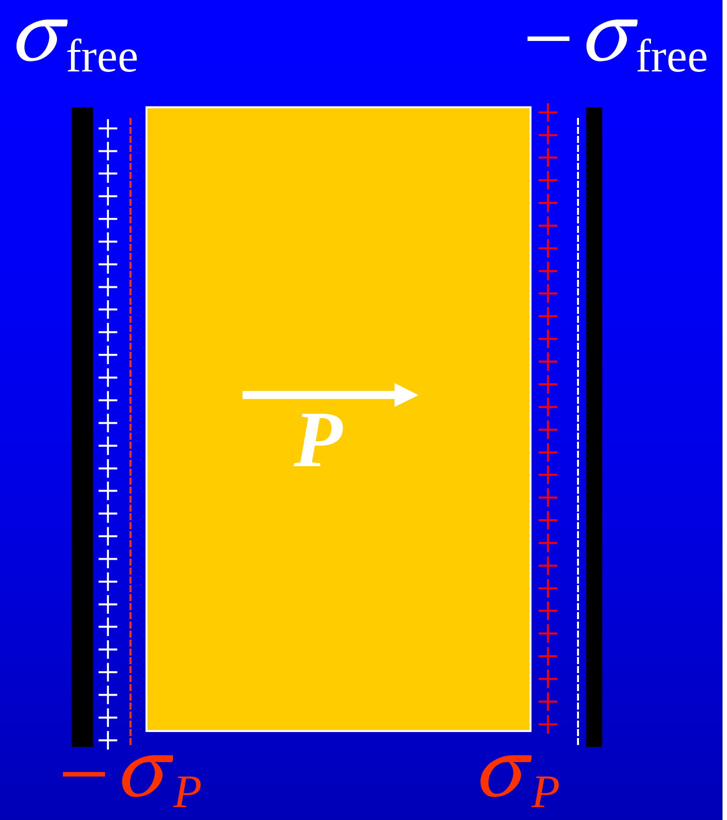
$$E=\frac{\sigma_{\text{free}}-\sigma_P}{\varepsilon_0}=\frac{\sigma_{\text{free}}-P}{\varepsilon_0}.$$

Using $P=\chi_e\varepsilon_0 E$, we have

$$E=\frac{\sigma_{\text{free}}-\chi_e\varepsilon_0 E}{\varepsilon_0}.$$

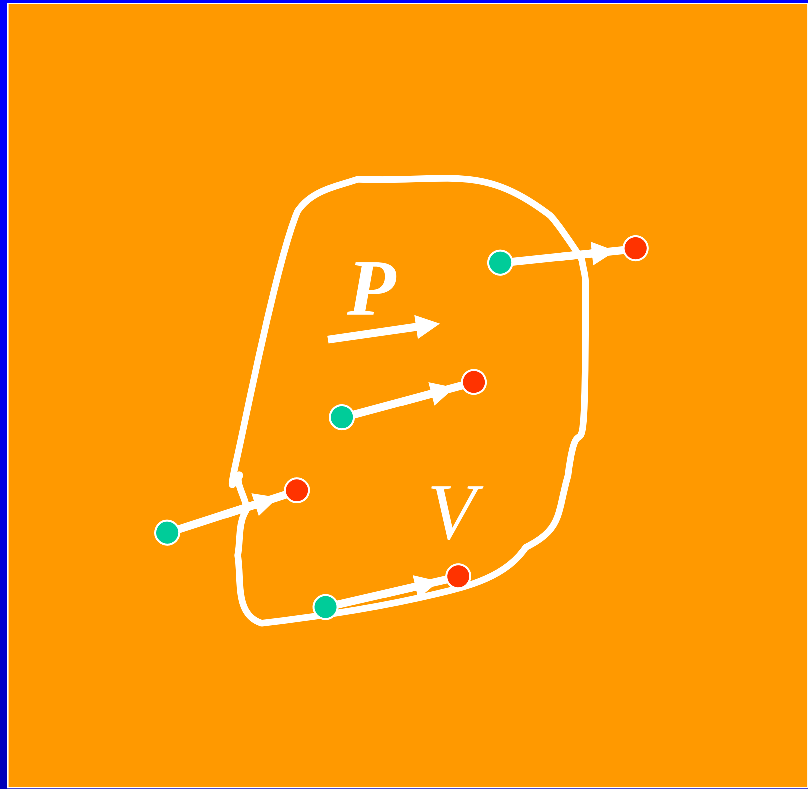
$$\Downarrow$$

$$E=\frac{\sigma_{\text{free}}}{\varepsilon_0(1+\chi_e)}.\qquad \Delta V=Ed=\frac{\sigma_{\text{free}}d}{\varepsilon_0(1+\chi_e)}=\frac{Qd}{\varepsilon_0(1+\chi_e)S}.$$



$$C_d=\frac{\varepsilon_0(1+\chi_e)S}{d}=(1+\chi_e)C. \Longrightarrow \boxed{\varepsilon_r=1+\chi_e.}$$

How many charge is gained or lost in a volume V due to the polarization?



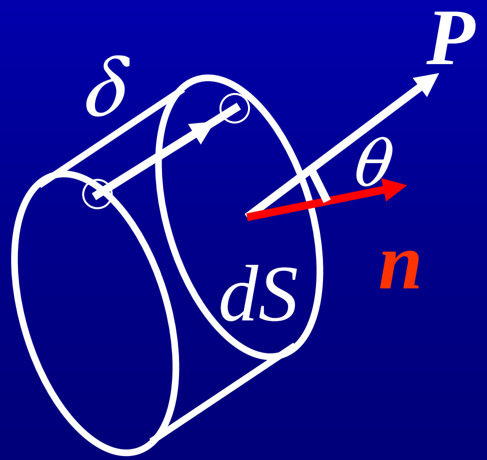
The charge moved across a surface element dS is

$$nq\delta dS\cos\theta = \boldsymbol{P}\cdot\boldsymbol{n}dS.$$

The total charge displaced out of the volume V is

$$\oint_S \boldsymbol{P}\cdot\boldsymbol{n}dS.$$

The total charge left behind the volume is



$$q_P = -\oint_S \boldsymbol{P}\cdot\boldsymbol{n}dS = -\oint_S \boldsymbol{P}\cdot d\boldsymbol{S}.$$

From Gauss' law , we have

$$\varepsilon_0 \oint_S \boldsymbol{E} \cdot d\boldsymbol{S} = q_{\text{total}} = q_{\text{free}} + q_P$$

$$= q_{\text{free}} - \oint_S \boldsymbol{P} \cdot d\boldsymbol{S},$$

or

$$\oint_S (\varepsilon_0 \boldsymbol{E} + \boldsymbol{P}) \cdot d\boldsymbol{S} = q_{\text{free}}.$$

electric displacement (vector)

We define $\boldsymbol{D} = \varepsilon_0 \boldsymbol{E} + \boldsymbol{P}.$

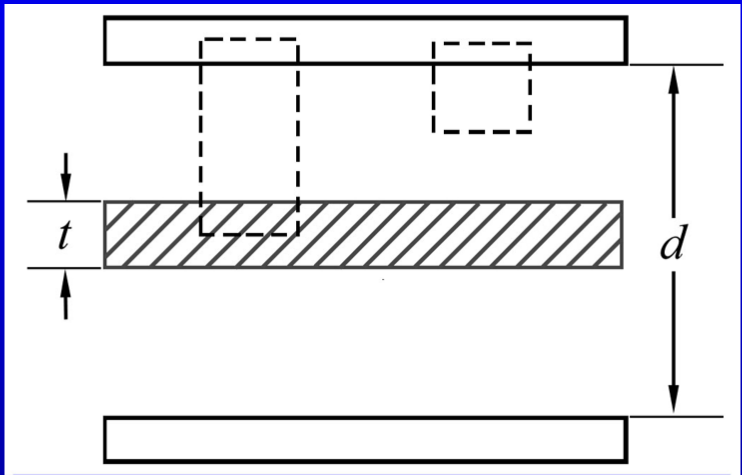
Therefore, Gauss' law can be rewritten as

$$\boxed{\oint_S \boldsymbol{D} \cdot d\boldsymbol{S} = q_{\text{free}}.}$$

For isotropic dielectrics, we have

$$\boldsymbol{D} = \varepsilon_0 \boldsymbol{E} + \boldsymbol{P} = \varepsilon_0 (1 + \chi_e) \boldsymbol{E} = \varepsilon_0 \varepsilon_r \boldsymbol{E} = \varepsilon \boldsymbol{E}.$$

Example 16.2 Calculate the capacitance of a parallel --plate capacitor with dielectric slab inserted midway between two plates.



Solution: Making a cylindrical gaussian surface with one end inside one of the metal plates and the other base inside the slab of dielectrics or inside the the gaps. With the use of Gauss's law, we have

$$DS' = \sigma_{\text{free}} S' \quad \text{or} \quad D = \sigma_{\text{free}} = \frac{Q}{S},$$

where *S'* the area of the base and S in the area of the plate.. Thus the electric field is

$$E = \begin{cases} \dfrac{D}{\varepsilon_0} = \dfrac{Q}{\varepsilon_0 S}, & \text{inside the gap,} \\ \dfrac{D}{\varepsilon_r \varepsilon_0} = \dfrac{Q}{\varepsilon_r \varepsilon_0 S}, & \text{inside the dielectrics.} \end{cases}$$

The potential difference between two plates is

$$\Delta V = \frac{Q}{\varepsilon_0 S}(d - t) + \frac{Q}{\varepsilon_r \varepsilon_0 S} t.$$

Therefore, the capacitance is

$$C = \frac{Q}{\Delta V} = \frac{\varepsilon_0 S}{(d-t) - t/\varepsilon_r} = \frac{\varepsilon_0 S}{d - (1 - 1/\varepsilon_r)t}.$$

When $t=d$, we have $$C = \frac{\varepsilon_r \varepsilon_0 S}{d} = \varepsilon_r C_0.$$

When $t=0$, we have $$C == \frac{\varepsilon_0 S}{d} = C_0.$$

The boundary conditions

n ---normal vector

$1 \to 2$

$$\boldsymbol{D}_2 \cdot \boldsymbol{n}S + \boldsymbol{D}_1 \cdot (-\boldsymbol{n})S = \sigma_{\text{free}} S$$

$$\Downarrow$$

$$D_{2n} - D_{1n} = \sigma_{\text{free}},$$

or $$\frac{E_{2n}}{\varepsilon_2} - \frac{E_{1n}}{\varepsilon_1} = \sigma_{\text{free}}.$$

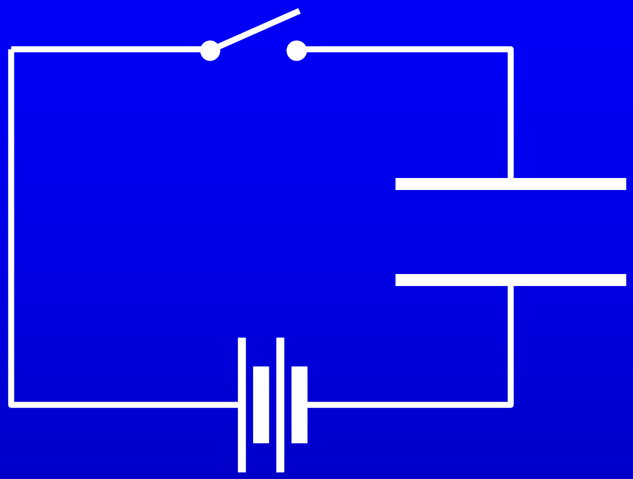
the boundary condition
for the normal component

$$\oint \boldsymbol{E} \cdot d\boldsymbol{l} = 0. \Longrightarrow E_{1t} = E_{2t}.$$

the boundary condition
for the tangential component

The energy density in the dielectrics

The energy storage of a charged capacitor



The work required to transfer a charge dq from the low plate to the upper plate is

$$dW = \Delta V dq = \frac{q}{C} dq.$$

Then, the total work required to increase the charge from zero to some amount Q is

$$W = \int_0^Q \frac{q}{C} dq = \frac{1}{2} \frac{Q^2}{C}.$$

Therefore, the electric energy potential in the capacitor is

$$U = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}C(\Delta V)^2.$$

For a parallel-plate capacitor ,

$$C = \frac{\varepsilon S}{d} \qquad \text{and} \qquad \Delta V = Ed,$$

Then

$$U = \frac{1}{2}\frac{\varepsilon S}{d}\left(Ed\right)^2 = \frac{1}{2}\varepsilon E^2(Sd).$$

The energy density is

$$u = \frac{1}{2}\varepsilon E^2 = \frac{1}{2}\boldsymbol{E}\cdot\boldsymbol{D}.$$

Assignment: 16.9; 16.12

$$q_P = -\oint_S \boldsymbol{P} \cdot \boldsymbol{n} dS = -\oint_S \boldsymbol{P} \cdot d\boldsymbol{S} = -\int_V \nabla \cdot \boldsymbol{P} d\tau.$$

$$q_P = \int_V \rho_P d\tau.$$

polarization charge density

For an arbitrary volume,

$$\int_V \rho_P d\tau = -\int_V \nabla \cdot \boldsymbol{P} d\tau,$$

then $\rho_P = -\nabla \cdot \boldsymbol{P}.$